

Hyodae Seo

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BIO

Dr. Seo is a physical oceanographer and climate scientist, with research and teaching interests in the coupled interactions between the ocean and atmosphere that shape weather and climate. His work integrates high-resolution coupled modeling, geophysical fluid dynamics, and analyses of in situ and satellite observations to advance understanding of ocean–atmosphere boundary layer processes, improve their representation in numerical models, assess their influence on ocean circulation, weather systems, and climate variability, and contributes to the development of ocean observing and modeling strategies with applications to offshore wind energy, extreme weather prediction, and climate resilience.

EDUCATION

- 2007 **Ph.D., Oceanography**, *Scripps Institution of Oceanography, UC San Diego*
 Thesis title: [Mesoscale coupled ocean-atmosphere interaction](#)
 2002 **B.S., Meteorology**, *Yonsei University, Seoul, South Korea*

PROFESSIONAL APPOINTMENTS

- 2024/08 – present **Associate Professor**, *Department of Oceanography, UHM*
 2023/08 – present **Senior Scientist**, *Physical Oceanography Department, WHOI*
 2018/07 – 2023/07 **Associate Scientist with Tenure**, *Physical Oceanography Department, WHOI*
 2014/07 – 2018/06 **Associate Scientist**, *Physical Oceanography Department, WHOI*
 2010/10 – 2014/06 **Assistant Scientist**, *Physical Oceanography Department, WHOI*
 2010/01 – 2010/09 **Visiting Assistant Researcher**, *International Pacific Research Center, UHM*
 2008/09 – 2009/12 **NOAA Climate and Global Change Postdoctoral Fellow**, *IPRC, SOEST, UHM*
 2008/01 – 2008/08 **NOAA Climate and Global Change Postdoctoral Fellow**, *UCLA*
 2007/07 – 2007/12 **Visiting Scientist**, *IPRC, SOEST, UHM*
 2002/09 – 2007/06 **Graduate Research Assistant**, *Scripps Institution of Oceanography, UCSD*

AWARDS AND HONORS

- WHOI Francis E. Fowler IV Ocean and Climate Fellow (2022–2023, [link](#))
- Office of Naval Research (ONR) Young Investigator Award (2015–2019)
- NOAA Climate and Global Change Postdoctoral Fellowship (2007–2009, [Class 17](#))
- Physical Oceanography Dissertation Symposium V, Honolulu (2008)
- NCAR Advanced Study Program (ASP) Graduate Student Visiting Fellowship (2006)
- Scripps Institution of Oceanography Frieman Director's Prize for Excellence in Graduate Student Research (2006, [UCSD press release](#))
- Outstanding Student Paper Award, Ocean Sciences Meeting, Honolulu (2006)

TEACHING AND ADVISING

@ UHM (since August 2024)

- **Spring 2025, OCN760:** Topics in Physical Oceanography, Air-Sea Interaction & Offshore Wind
- **Spring 2025, OCN780:** Ocean Seminar
- **Fall 2024, GES 311:** The Changing Earth and Climate System. Guest lecture, “Why Offshore Wind: Necessity, Challenges and Oceanographers’ Role”, Nov 21, 2024

@ MIT/WHOI Joint Program (until July 2024)

- **Spring 2023, 12.870:** Air–Sea Interaction and Boundary Layers. MIT/WHOI Joint Program Course. Co-taught with Tom Farrar
- **Fall 2017, 2019, 2021, 12.860:** Climate Variability and Diagnostics. MIT/WHOI Joint Program Course. Co-taught with Caroline Ummenhofer
- **Fall 2016, 2017, 2018, 2019, 12.808:** Introduction to (Observational) Physical Oceanography, MIT/WHOI Joint Program Course. Co-taught with John Toole

ADVISING AND MENTORING

@ UHM (since August 2024)

- Ph.D. Thesis Committee, Margaret Bradley, Oceanography Department, UHM (2024–present)
- Ph.D. Thesis Committee, Ryo Dobashi, Oceanography Department, UHM (2024–present)
- M.S. Thesis Committee, Yuta Norden, Oceanography Department, UHM (2025–present)
- Ph.D. Thesis Committee, Ajin Cho, Yonsei University, South Korea (2025–present)
- Comprehensive Exam Committee, Carla Balzeau, Oceanography Department, UHM (2025)
- Comprehensive Exam Committee, Vera Stockmayer, Oceanography Department, UHM (2025)

@ MIT/WHOI Joint Program (until July 2024)

- Ph.D. General Exam Advisor, Anthony Meza, MIT-WHOI Joint Program (2022)
- Ph.D. Thesis Committee, Hanyuan Liu, MIT-WHOI Joint Program (2021–2024)
- Ph.D. External Examiner, Christoph Renkl, Dalhousie University, Oceanography (2020)
- Ph.D. Thesis Committee, Jared Buckley, University of Massachusetts, Dartmouth (2015– 2019)
- M.S. Thesis Committee, Xin Zhou, Florida Institute of Technology (2015– 2016)
- M.S. Thesis committee, Sara Bosshart, MIT-WHOI Joint Program, 2012– 2013

Graduate Advisor

- Rachel Wyn Pauly: Ph.D. student, Oceanography Department, UHM (2025–present)
- Clint Reyes: Summer graduate student, Dept. Ocean and Resources Engineering, UHM (2025)

Research Advisor

@ UHM (since August 2024)

- Dr. César Sauvage: Oceanographic Researcher (2024/11–present)
- Dr. Sid Kerhalkhar: Postdoc (2025/09–present)

@ WHOI (until July 2024)

- Dr. Shikhar Rai, Postdoc (2023–present)
- Dr. Benjamin Barr, Postdoc (2023–present)
- Christoph Renkl, Postdoc (2023–2025, Now Interim Professor at the Univ. of Bonn)
- Dr. Alma Carolina Castillo Trujillo, Postdoc (2021–2024, Now @ atdepth MRV)
- César Sauvage (2020–2025, now at UHM)
- John Steffen (2019–2021, now at National Hurricane Center/NOAA)
- Rhys Parfitt (2016–2018, now Associate Professor, Florida State University)

Visiting Scholars and Guest Students

@ UHM (since August 2024)

- Prof. Changhyun Yoo: Professor, Ewha Womans University, South Korea (2025–2026)

@ WHOI (until July 2024)

- Yoo-Jun Kim: University of Tokyo (2023/11–2024/03)
- Peisen Tan: University of Miami (Summer of 2023)
- Sam Bartusek: Princeton University (WHOI Summer Student Fellow 2019; [Bartusek et al. 2021](#))

- Yuqing Liu: Bremen University (2018)
- Manthan Shah: Northeastern University (2018)
- Xu Yuan: University of Twente (2018; [Yuan et al. 2020](#))
- Xiaolin Jin: Tsinghua University (2017; [Jin et al. 2018a](#); [Jin et al. 2018b](#))
- Samuel Coakley: Rutgers University (WHOI Summer Student Fellow 2017)
- David Smith: Northeastern University (2016)
- Channing Prend: Columbia University (WHOI SSF 2016; [Prend et al. 2018](#); [Weller et al. 2019](#))
- Siraput Jongaramrungruang: U. Cambridge (WHOI SSF 2015; [Jongaramrungruang et al. 2017](#))
- Alicia Camacho: Valparaíso University (UCAR SOARS student 2015; [Kwon et al. 2018](#))
- Carlos Martinez: Texas A&M University (UCAR SOARS student 2014; [Kwon et al. 2018](#))

PROFESSIONAL ACTIVITIES

Community Activities and Services

- UCAR President's Advisory Committee on University Relations (PACUR): 2023–present ([link](#))
- WHOI Representative of UCAR (2018–2023)
- Co-Chair: US CLIVAR Working Group on Mesoscale and Frontal–Scale Ocean–Atmosphere Interactions and Influence on Large-scale Climate (2019–2025, [link](#))
- Member US CLIVAR Air–Sea Transition Zone Study Group (2022–2024, [link](#))
- Associate Editor: *Journal of Climate* (2019–present)
- Member: AMS STAC Committee on Air–Sea Interaction (2019–2022)
- Member: US CLIVAR Process Study and Model Improvement Panel (2016–2020)
- Associate Editor: *Atmospheric Science Letters* (2015–2017)
- Member: AMS STAC Committee on Coastal Environment (2011–2017)

Committee and Service

@ UHM (since August 2024)

- Department of Oceanography Awards Committee (2025–present)

@ WHOI (until July 2024)

- WHOI Physical Oceanography Recruitment Committee for Scientific Staff (2020–2021)
- WHOI High-Performance Computing Strategy Committee (2022–2023)
- WHOI Summer Student Fellowship Coordinator (2017–2019)
- WHOI High-Performance Computing User Advisory Committee (2010–2017)
- WHOI Physical Oceanography Department Seminar Series Organizer (2010–2011)

Reviewer:

- Papers

Atmospheric Science Letters, Bulletin of the American Meteorological Society, Climate Dynamics, Communications Earth & Environment, Environmental Research Letters, Geophysical Research Letters, Journal of Advances in Modeling Earth Systems, Journal of Atmospheric Sciences, Journal of Climate, Journal of Marine Systems, Journal of Physical Oceanography, Journal of Geophysical Research-Atmospheres, Journal of Geophysical Research- Oceans, Monthly Weather Review, Nature, Nature Communications, npj Climate and Atmospheric Science, Ocean Dynamics, Ocean Modelling, Remote Sensing, Science Advances, Scientific Reports, etc.

- Proposals

NSF Physical Oceanography, NSF Climate and Large-Scale Dynamics, NSF Office of Polar Programs, NSF United States–Israel Binational Science Foundation, India Monsoon Missions I–III, Chile National Research Funding FONDECYT, CSCS Swiss National Supercomputing Centre

Agency proposal review panels

hyode@hawaii.edu

- NASA SWOT (2012); DOE ESMD (2020)

Society Memberships

- American Geophysical Union (2002–present)
- American Meteorological Society (2002–present)
- The Oceanography Society (2012–present)

PUBLICATIONS

Summary: 67 published or in press & 4 Under review

Total citations: 3405 (Google Scholar); 2602 (Web of Science)

H-index: 32 (Google Scholar), 25 (Web of Science)

* denotes the work led by a student and a postdoc under my direct supervision

denotes publication based on work done at UH

The contributions listed are from the terms described in <https://credit.niso.org/>.

Under Review or Revision:

- #71. ***Rai, S.**, S. Wijffels, **H. Seo**, J. T. Farrar, T. Lee, and R. A. Weller, Sampling Scale Mismatch Drive Discrepancies between Satellite and TAO Buoy Measurements of Ocean Surface Winds *J. Geophys. Res. Oceans*, Submitted
 Conceptualization, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resource, Supervision, Writing-original draft, Writing-review & editing
- #70. Yang, S., H.-R. Kim, **B. W. Barr**, C. C. Nam, **H. Seo**, M. Bourassa, S. Cocke, M. Kwon, D.-H. Cha, M. Park, D.-S. R. Park, H.-J. Bae, B.-M. Kim, Wave Tank-Inspired Sea Spray Parameterization Enhances Typhoon Intensity Forecasts. *Science Advances*, Submitted
 Supervision, Writing-review & editing.
- #69. Cho, A., I.-J. Moon, **H. Seo**, R. Sun, M. R. Mazloff, A. C. Subramanian, B. D. Cornuelle, A. J. Miller, and H. Song: Modulation of Tropical Cyclone Intensity by Current-Wind Interaction, *npj Climate and Atmospheric Science*, Under Minor Revision
 Supervision, Writing-review & editing.
- #68. *Sauvage, C., **H. Seo**, S. Zippel, C.-A. Clayson, and J. B. Edson: Fetch-dependent Surface Wave Responses to Offshore Wind Farms in the Northeast U.S. Coast, *Journal of Geophysical Research-Oceans*, Minor Revision Submitted.
 Conceptualization, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resource, Supervision, Writing-original draft, Writing-review & editing

Published:

2025

- #67. **Seo, H.**, C. Sauvage, C. Renkl, J. K. Lundquist, and A. Kirincich, 2025: Sea Surface Warming and Ocean-to-Atmosphere Feedback Driven by Large-Scale Offshore Wind Farms under Seasonally Stratified Conditions. *Science Advances*, 11, eadw7603,
<https://www.science.org/doi/10.1126/sciadv.adw7603>
 Conceptualization, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resource, Supervision, Writing-original draft, Writing-review & editing
- #66. Raj, A., B. P. Kumar, V. Jampana, P. G. Remya, **H. Seo**, N. Sureshkumar, and P. R. Rao, 2025: Wind-Stress Misalignment in the Presence of Swells in the Bay of Bengal. *Frontiers in Marine Science*, 12. <https://doi.org/10.3389/fmars.2025.1647849>
 Conceptualization, Investigation, Methodology, Writing-original draft, Writing-review & editing
- #65. Yang, S., H. Bae, M.-S. Park, M. Bourassa, C. C. Nam, S. Cocke, D. W. Shin, B. W. Barr, **H. Seo**, D.-H. Cha, M.-H. Kwon, D. Kim, K.-Y. Jung, and B.-M. Kim, 2025: Sea spray effects on typhoon prediction in the Yellow and East China Seas: Case studies using a coupled atmosphere-ocean-wave

model for Lingling (2019) and Maysak (2020). *Environmental Research Letters*, 20, 054028.

<https://doi.org/10.1088/1748-9326/adc616>

Conceptualization, Investigation, Writing-original draft, Writing-review & editing

- #64. Cho, A., H. Song, **H. Seo**, R. Sun, M. R. Mazloff, A. C. Subramanian, B. D. Cornuelle, and A. J. Miller, 2025: Dynamic and Thermodynamic coupling between the Atmosphere and Ocean near the Kuroshio Current and Extension System. *Ocean Modelling*, 194, 102496. <https://doi.org/10.1016/j.ocemod.2024.102496>
Conceptualization, Investigation, Supervision, Writing-original draft, Writing-review & editing

2024

63. *Sauvage, C., **H. Seo**, B. W. Barr, J. B. Edison, and C. A. Clayson, 2024: Misaligned Wind-Waves Behind Atmospheric Cold Fronts. *Journal of Geophysical Research-Oceans*, 129, e2024JC021162. <https://doi.org/10.1029/2024JC021162>
Conceptualization, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resource, Supervision, Writing-original draft, Writing-review & editing
62. Shinoda, T., T. G. Jensen, Z. Lachkar, Y. Masumoto, and **H. Seo**, 2024: Modeling the Indian Ocean, Intraseasonal variability in the Indian Ocean region. *The Indian Ocean and its role in the global climate system*. Elsevier Publishing, Editors: C. C. Ummerhofer and R. R. Hood. <https://doi.org/10.1016/B978-0-12-822698-8.00013>
Conceptualization, Investigation, Supervision, Writing-original draft, Writing-review & editing

2023

61. *Steffen, J. D., **H. Seo**, C. A. Clayson, S. Pei, and T. Shionda, 2023: Impacts of Tidal Mixing on Diurnal to Intraseasonal Air-Sea Interactions in the Maritime Continent. *Deep-Sea Research Part II*, 212, 105343. <https://doi.org/10.1016/j.dsr2.2023.105343>
Conceptualization, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resource, Supervision, Writing-original draft, Writing-review & editing
60. *Castillo-Trujillo A. C., Y.-O. Kwon, P. Fratantoni, K. Chen, **H. Seo**, M. A. Alexander, and V. S. Saba, 2023: An evaluation of eight global ocean reanalyses for the Northeast U.S. continental shelf. *Prog. Oceanogr.*, <https://doi.org/10.1016/j.pocean.2023.103126>
Funding acquisition, Supervision, Writing-review & editing
59. Clayson, C. A., C. A. DeMott, S. P. de Szoeke, P. Chang, G. R. Foltz, R. Krishnamurthy, T. Lee, A. Molod, D. G. Ortiz-Suslow, J. Pullen, D. H. Richter, **H. Seo**, P. C. Taylor, E. Thompson, B. V. Bôas, C. J. Zappa, and P. Zuidema, 2023: A New Paradigm for Observing and Modeling of Air-Sea Interactions to Advance Earth System Prediction. A US CLIVAR Report, US CLIVAR Project Office, 86 pp. <https://doi.org/10.5065/24j7-w583>
Conceptualization, Formal Analysis, Investigation, Methodology, Writing-original draft, Writing-review & editing
58. Beaudin, E., E. Di Lorenzo, A. J. Miller, **H. Seo**, and Y. Joh, 2023: Impact of Extratropical Northeast Pacific SST on U.S. West Coast Precipitation, *Geophysical Research Letters*, 50, e2022GL102354. <https://doi.org/10.1029/2022GL102354>
Conceptualization, Supervision, Methodology, Resource, Writing-review & editing
57. *Sauvage, C., **H. Seo**, C. A. Clayson, and J. B. Edson, 2023: Improving wave-based air-sea momentum flux parameterization in mixed seas. *Journal of Geophysical Research-Oceans*, 128, e2022JC019277. <https://doi.org/10.1029/2022JC019277>
Conceptualization, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resource, Supervision, Writing-review & editing
56. **Seo, H.**, L. W. O'Neill, M. A. Bourassa, A. Czaja, K. Drushka, B. Fox-Kemper, I. Frenger, S. T. Gille, B. P. Kirtman, S. Minobe, A. G. Pendergrass, L. Renault, M. J. Roberts, N. Schneider, R. J. Small, A. Stoffelen, and Q. Wang, 2023: Ocean Mesoscale and Frontal-scale Ocean-Atmosphere

Interactions and Influence on Large-scale Climate: A Review. *Journal of Climate*, 36, 1981-2013.
<https://doi.org/10.1175/JCLI-D-21-0982.1>
 Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Visualization,
 Writing-original draft, Writing-review & editing

2021

55. Kwak, K. H. Song, J. Marshall, **H. Seo**, and D. McGillicuddy, Jr., 2021: Suppressed pCO₂ in the Southern Ocean due to the interaction between current and wind. *Journal of Geophysical Research-Oceans*, 126, e2021JC017884. <https://doi.org/10.1029/2021JC017884>
 Conceptualization, Supervision, Writing-original draft, Writing-review & editing
54. Shinoda, T., S. Pei, W. Wang, J. X. Fu, R.-C. Lien, **H. Seo**, and A. Soloviev, 2021: Climate Process Team: improvement of ocean component of NOAA Climate Forecast System relevant to Madden-Julian Oscillation simulations. *Journal of Advances in Modeling Earth Systems*, 13, e2021MS002658. <https://doi.org/10.1029/2021MS002658>
 Conceptualization, Investigation, Supervision, Writing-original draft, Writing-review & editing
53. **Seo, H.**, H. Song, L. W. O'Neill, M. R. Mazloff, and B. D. Cornuelle, 2021: Impacts of ocean currents on the South Indian Ocean extratropical storm track through the relative wind effect. *Journal of Climate*, 34, 9093-9113. <https://doi.org/10.1175/JCLI-D-21-0142>
 Conceptualization, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resource, Writing-original draft, Writing-review & editing
52. Pei, S., T. Shinoda, J. Steffen, and **H. Seo**, 2021: Substantial Sea Surface Temperature cooling in the Banda Sea associated with the Madden-Julian Oscillation in the boreal winter of 2015. *Journal of Geophysical Research-Oceans*, 126, e2021JC017226. <https://doi.org/10.1029/2021JC017226>
 Conceptualization, Investigation, Writing-review & editing
51. Shroyer, E., and Co-authors, including **H. Seo**, 2021: Bay of Bengal Intraseasonal Oscillation and the 2018 Monsoon Onset. *Bull. Amer. Meteor. Soc.*, 102(10), E1936-E1951.
<https://doi.org/10.1175/BAMS-D-20-0113>
 Conceptualization, Investigation, Writing-original draft, Writing-review & editing
50. *Bartusek, S., **H. Seo**, C. C. Ummerhofer, and J. D. Steffen, 2021: The role of nearshore air-sea interactions for landfalling atmospheric rivers on the U.S. West Coast. *Geophysical Research Letters*, 48, e2020GL091388. <https://doi.org/10.1029/2020GL091388>
 Conceptualization, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resource, Supervision, Writing-review & editing
49. Sun, R., A. C. Subramanian, B. D. Cornuelle, M. R. Mazloff, A. J. Miller, F. M. Ralph, **H. Seo**, and I. Hoteit, 2021: The role of air-sea interactions in atmospheric river events: Case studies using the SKRIPS regional coupled model. *Journal of Geophysical Research-Atmosphere*, 126, e2020JD032885. <https://doi.org/10.1029/2020JD032885>
 Conceptualization, Writing-review & editing

2020

48. Liang, Y.-C., M.-H. Lo, C.-W. Lan, **H. Seo**, C. C. Ummerhofer, S. Yeager, R.-J. Wu, and J. D. Steffen, 2020: Amplified Seasonal Cycle in Hydroclimate over the Amazon River Basin and its Plume Region in the Atlantic. *Nat. Commun.*, 11, 4390. <https://doi.org/10.1038/s41467-020-18187-0>
 Conceptualization, Supervision, Formal Analysis, Investigation, Writing-review & editing
47. *Yuan, X., C. C. Ummerhofer, **H. Seo**, and Z. Su, 2020: Relative contributions of heat flux and wind stress on the spatiotemporal upper-ocean variability in the tropical Indian Ocean. *Environmental Research Letters*, 15, 084047. <https://doi.org/10.1088/1748-9326/ab9f7f>
 Conceptualization, Formal Analysis, Methodology, Resource, Supervision, Writing-original draft, Writing-review & editing
46. Gentemann, C., C. A. Clayson, S. Brown, T. Lee, R. Parfitt, J. T. Farrar, M. Bourassa, P. J. Minnett, **H. Seo**, S. T. Gille, and V. Zlotnicki, 2020: FluxSat: Measuring the ocean-atmosphere turbulent
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- exchange of heat and moisture from space. *Remote Sensing*, 12, 1796.
<https://doi.org/10.3390/rs12111796>
 Conceptualization, Formal Analysis, Investigation, Validation, Writing-review & editing
45. Sprintall, J., K. A. Reed, A. H. Butler, G. R. Foltz, S. G. Penny, and **H. Seo**, 2020: Best Practice Strategies for Process Studies designed to Improve Climate Modelling, *Bulletin of the American Meteorological Society*, 101, E1842-E1850. <https://doi.org/10.1175/BAMS-D-19-0263>.
 Conceptualization, Writing-review & editing
 44. Song, H., J. Marshall, D. J. McGillicuddy Jr., and **H. Seo**, 2020: The impact of the current-wind interaction on the vertical processes in the Southern Ocean. *Journal of Geophysical Research, Oceans*, 125, e2020JC016046. <https://doi.org/10.1029/2020JC016046>
 Conceptualization, Methodology, Formal Analysis, Supervision, Writing-review & editing
 43. Gopalakrishnan, G., A. C. Subramanian, A. J. Miller, **H. Seo**, and D. Sengupta, 2020: Estimation and prediction of the upper ocean circulation in the Bay of Bengal, *Deep-Sea Research Part II*, 172, 104721. <https://doi.org/10.1016/j.dsr2.2019.104721>
 Conceptualization, Writing-review & editing
 42. Hagos, S., G. R. Foltz, C. Zhang, E. Thompson, **H. Seo**, S. Chen, A. Capotondi, K. A. Reed, C. DeMott, and A. Protat, 2020: Atmospheric Convection and Air-Sea Interactions over the Tropical Oceans: Scientific Progress, Challenges and Opportunities. *Bulletin of the American Meteorological Society*, 101, E253–E258. <https://doi.org/10.1175/BAMS-D-19-026>
 Conceptualization, Writing-review & editing
 41. Kwon, Y.-O., **H. Seo**, C. C. Ummenhofer, and T. M. Joyce, 2020: Impact of Multidecadal Variability in Atlantic SST on Winter Atmospheric Blocking. *Journal of Climate*, 33, 867-892. <https://doi.org/10.1175/JCLI-D-19-0324>
 Conceptualization, Investigation, Writing-review & editing
- 2019**
40. **Seo, H.**, A. C. Subramanian, H. Song, and J. S. Chowdary, 2019: Coupled effects of ocean current on wind stress in the Bay of Bengal: Eddy energetics and upper ocean stratification. *Deep-Sea Research Part II*, 168, 104617. <https://doi.org/10.1016/j.dsr2.2018.09.007>
 Conceptualization, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resource, Supervision, Writing-original draft, Writing-review & editing
 39. *Prend, C. J., **H. Seo**, R. A. Weller, and J. T. Farrar, 2019: Impact of freshwater plumes on intraseasonal upper ocean variability in the Bay of Bengal. *Deep-Sea Research Part II*, 161, 63-71. <https://doi.org/10.1029/2018GL0810>
 Conceptualization, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resource, Supervision, Writing-review & editing
 38. Joyce, T. M., Y.-O. Kwon, **H. Seo**, and C. C. Ummenhofer, 2019: Meridional Gulf Stream shifts can influence wintertime variability in the North Atlantic Storm Track and Greenland Blocking. *Geophysical Research Letters*, 46, 1702-1708. <https://doi.org/10.1175/JCLI-D-18-0413.1>
 Conceptualization, Writing-review & editing
 37. Weller, R. A., J. T. Farrar, **H. Seo**, C. J. Prend*, D. Sengupta, J. Sree Lekha, M. Ravichandran, and R. Venkatesen, 2019: Moored observations of the surface meteorology and air-sea fluxes in the northern Bay of Bengal in 2015, *Journal of Climate*, 32, 549-573. <https://doi.org/10.1016/j.dsr2.2019.07.00>
 Conceptualization, Writing-review & editing
- 2018**
36. *Parfitt, R., and **H. Seo**, 2018: A New Framework for Near-Surface Wind Convergence over the Kuroshio Extension and Gulf Stream in Wintertime: The Role of Atmospheric Fronts. *Geophysical Research Letters*, 45, 9909–9918. <https://doi.org/10.1175/JCLI-D-18-0413.1>

- Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resource, Supervision, Writing-review & editing
35. Pullen, J., R. Allard, **H. Seo**, A. J. Miller, S. Chen, L. P. Pezzi, T. Smith, P. Chu, J. Alves, and R. Caldeira, 2018: Coupled ocean-atmosphere forecasting at short and medium time scales. *Journal of Marine Research*, 75, 877-921. https://elischolar.library.yale.edu/journal_of_marine_research/4
Conceptualization, Investigation, Writing-original draft, Writing-review & editing
 34. *Jin, X, Y.-O. Kwon, C. C. Ummerhofer, **H. Seo**, Y. Kosaka, and J. S. Wright, 2018: Distinct mechanisms of decadal subsurface heat content variations in the eastern and western Indian Ocean modulated by tropical Pacific SST, *Journal of Climate*, 31, 7751-7769.
<https://doi.org/10.1175/JCLI-D-18-0184.1>
Investigation, Methodology, Supervision, Writing-review & editing
 33. *Jin, X., Y.-O. Kwon, C. C. Ummerhofer, **H. Seo**, F. U. Schwarzkopf, A. Biastoch, C. W. Böning, and J. S. Wright, 2018: Influences of Pacific Climate Variability on Decadal Subsurface Ocean Heat Content Variations in the Indian Ocean. *Journal of Climate*, 31, 4157-4174.
<https://doi.org/10.1175/JCLI-D-17-0654.1>
Investigation, Methodology, Supervision, Writing-review & editing
 32. Kwon, Y.-O., A. Camacho, C. Martinez-Zayas, and **H. Seo**, 2018: North Atlantic Eddy-driven Jet and Blocking Variability in the Community Earth System Model Version 1 Large Ensemble Simulations. *Climate Dynamics*, 51, 3275-3289. <https://doi.org/10.1007/s00382-018-4078-6>
Conceptualization, Investigation, Writing-review & editing

2017

31. **Seo, H.**, Y.-O. Kwon, T. M. Joyce, and C. C. Ummerhofer, 2017: On the predominant nonlinear response of the extratropical atmosphere to meridional shift of the Gulf Stream. *Journal of Climate*, 30, 9679-9702. <https://doi.org/10.1175/JCLI-D-16-0707.1>
Conceptualization, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resource, Writing-review & editing
30. Morrow, R., L.-L. Fu, T. Farrar, **H. Seo**, P.-Y. Le Traon, 2017: Ocean eddies and mesoscale variability. Satellite Altimetry Over Oceans and Land Surfaces, In D. Stammer and A. Cazenave, editors, *Satellite Altimetry Over Ocean and Land Surfaces*. CRC Press, Taylor and Francis Group, [Satellite Altimetry Over Oceans and Land Surfaces](#)
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26. Miller, A. J., M. Collins, S. Gualdi, T. G. Jensen, V. Misra, L. P. Pezzi, D. W. Pierce, D. Putrasahan, **H. Seo**, and Y.-H. Tseng, 2017: Coupled Ocean-Atmosphere-Hydrology Modeling and Predictions, *J. Mar. Res.*, 75, 361-402.
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23. Chowdary, J. S., G. Srinivas, T. S. Fousiya, A. Parekh, C. Gnanaseelan, **H. Seo**, and J. A. MacKinnon, 2016: Representation of Bay of Bengal Upper-Ocean Salinity in General Circulation Models. *Oceanography*, 29, 38–49. <https://doi.org/10.5670/oceanog.2016.37>
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15. Subramanian, A., M. Jochum, A. J. Miller, R. Neale, **H. Seo**, D. Waliser, and R. Murtugudde, 2014: The MJO and Global warming: A study in CCSM4. *Climate Dynamics*, 42, 2019-2031. <https://doi.org/10.1007/s00382-013-1846-1>
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13. Putrasahan, D. A. J. Miller, and **H. Seo**, 2013: Regional coupled ocean-atmosphere downscaling in the Southeast Pacific: Impacts on upwelling, mesoscale air-sea fluxes, and ocean eddies. *Ocean Dynamics*, 63, 463-488. <https://doi.org/10.1007/s10236-013-0608-2>
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12. **Seo, H.**, and S.-P. Xie, 2013: Impact of ocean warm layer thickness on the intensity of hurricane Katrina in a regional coupled model. *Meteorology and Atmospheric Physics*, 122, 19-32. <https://doi.org/10.1007/s00703-013-0275-3>
Conceptualization, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resource, Supervision, Writing-review & editing
11. Putrasahan, D. A. J. Miller, and **H. Seo**, 2013: Isolating Mesoscale Coupled Ocean-Atmosphere Interactions in the Kuroshio Extension Region. *Dynamics of Atmospheres and Oceans*, 63, 60-78. <https://doi.org/10.1016/j.dynatmoce.2013.04.001>
Conceptualization, Investigation, Methodology, Supervision, Writing-review & editing

2012

10. **Seo, H.**, K. H. Brink, C. E. Dorman, D. Koracin, and C. A. Edwards, 2012: What determines the spatial pattern in summer upwelling trends on the U.S. West Coast? *Journal of Geophysical Research-Oceans*, 117, C08012. <https://doi.org/10.1029/2012JC008016>
Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resource, Supervision, Writing-review & editing
9. Alexander, M.A., **H. Seo**, S.-P. Xie, and J.D. Scott, 2012: ENSO's impact on the gap wind regions of the eastern tropical Pacific Ocean. *Journal of Climate*, 25, 3549-3565. <https://doi.org/10.1175/JCLI-D-11-00320.1>
Formal Analysis, Investigation, Writing-original draft, Writing-review & editing

2006-2011

8. **Seo, H.**, and S.-P. Xie, 2011: Response and Impact of Equatorial Ocean Dynamics and Tropical Instability Waves in the Tropical Atlantic under Global Warming: A regional coupled downscaling study. *Journal of Geophysical Research-Oceans*, 116, C03026. <https://doi.org/10.1029/2010JC006670>
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6. **Seo, H.**, R. Murtugudde, M. Jochum, and A. J. Miller, 2008: Modeling of Mesoscale Coupled Ocean-Atmosphere Interaction and its Feedback to Ocean in the Western Arabian Sea. *Ocean Modelling*, 25, 120-131. <https://doi.org/10.1016/j.ocemod.2008.07.003>

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5. **Seo, H.**, M. Jochum, R. Murtugudde, A. J. Miller, and J. O. Roads, 2008: Precipitation from African Easterly Waves in a Coupled Model of the Tropical Atlantic. *Journal of Climate*, 21, 1417-1431. <https://doi.org/10.1175/2007JCLI1906.1>
Conceptualization, Formal Analysis, Investigation, Methodology, Writing-original draft, Writing-review & editing
4. Small, R. J., S. de Szoeke, S. P. Xie, L. O'Neill, **H. Seo**, Q. Song, P. Cornillon, M. Spall, and S. Minobe, 2008: Air-Sea Interaction over Ocean Fronts and Eddies. *Dynamics of Atmospheres and Oceans*, 45, 274-319. <https://doi.org/10.1016/j.dynatmoce.2008.01.001>
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3. **Seo, H.**, M. Jochum, R. Murtugudde, A. J. Miller, and J. O. Roads, 2007: Feedback of Tropical Instability Wave - induced Atmospheric Variability onto the Ocean. *Journal of Climate*, 20, 5842-5855. <https://doi.org/10.1175/JCLI4330.1>
Conceptualization, Formal Analysis, Investigation, Methodology, Writing-original draft, Writing-review & editing
2. **Seo, H.**, A. J. Miller and J. O. Roads, 2007: The Scripps Coupled Ocean-Atmosphere Regional (SCOAR) model, with applications in the eastern Pacific sector. *Journal of Climate*, 20, 381-402. <https://doi.org/10.1175/JCLI4016.1>
Conceptualization, Formal Analysis, Investigation, Methodology, Writing-original draft, Writing-review & editing
1. **Seo, H.**, M. Jochum, R. Murtugudde and A. J. Miller, 2006: Effect of Ocean Mesoscale Variability on the Mean State of Tropical Atlantic Climate. *Geophysical Research Letters*, 33, L09606. <https://doi.org/10.1029/2005GL025651>
Conceptualization, Formal Analysis, Investigation, Methodology, Writing-original draft, Writing-review & editing

FUNDED RESEARCH

⁺ denotes Seo is the Lead or Sole PI

@ UHM (since August 2024): [Raised >\\$1.69M as lead PI](#)

- **+ONR (current, \$524,481)**: Improving Air-Sea Flux Parameterizations to Better Understand the Diabatic Effects of Ocean and Surface Waves in Atmospheric Rivers.
- **+ONR (current, \$632,333)**: Improving the model simulation of surface wave impacts on air-sea fluxes, turbulent boundary layers, and their impacts on Indian monsoons in the Arabian Sea.
- **+WHOI (NASA) Subcontract (current, \$314,183)**: Improving coupled atmosphere-ocean processes in NU-WRF for the simulation of coast-threatening extratropical cyclones in the northeastern US.
- **+WHOI (DOE) Subcontract (current, \$87,582)**: Improving High Resolution Offshore Wind Resource Assessments and Forecasts using Observations in the MA/RI Lease Areas.
- **+WHOI (NSF) Subcontract (current, \$139,693)**: Improving understanding of coupled impacts of oceans and waves on air-sea fluxes in the US Northeast Coast.

@ WHOI (until July 2024), External Grants: [Raised ~\\$6.36M as lead PI, contributed to \\$11.49M as co-PI](#)

- **#ONR (current, \$50,000)**: Improving the model simulation of surface wave impacts on air-sea fluxes, turbulent boundary layers, and their impacts on Indian monsoons in the Arabian Sea.

- **#NOAA (current, \$773,753):** Exploiting coupled ocean-atmosphere-wave model simulations to identify observational requirements for air-sea interaction studies across the tropical Pacific. Co-PI: Wijffels
- **#NSF (current, \$831,828):** Improving understanding of coupled impacts of oceans and waves on air-sea fluxes in the US Northeast Coast. Co-PI: Sauvage, Clayson, and Edson. (\$139,693 subcontracted to UHM).
- **#NASA (current, \$1,166,106):** Improving coupled atmosphere-ocean processes in NU-WRF for the simulation of coast-threatening extratropical cyclones in the northeastern US. (\$314,183 subcontract to UHM).
- **DOE (current, \$10M):** Improving High Resolution Offshore Wind Resource Assessments and Forecasts using Observations in the MA/RI Lease Areas. co-PI, PI: Kirincich (WHOI). (\$87,582 subcontracted to UHM)
- **#NSF (current, \$497,273):** Collaborative Research: Coupled Ocean-Atmosphere Feedbacks Affecting California Coastal Climate: Current Conditions and Future Projections.
- **NOAA (current, \$658,442):** Regional multi-year prediction for the Northeast U.S. Continental Shelf. Co-PI. PI: Kwon (WHOI).
- **#NOAA (2019-2024, \$767,746):** Coupled Ocean-atmosphere interaction mediated by ocean mesoscale eddies in the Northwestern Tropical Atlantic Ocean. Co-PI: Clayson (WHOI)
- **#ONR (2017-2023, \$505,185):** Coupled Air-sea Interactions in the Bay of Bengal: Dynamics and Predictability of Monsoon Intraseasonal Oscillation.
- **NOAA (2017-2022, \$267,797):** Upper ocean processes in the Maritime Continent and their impact on the air-sea interaction and MJO predictability, co-PI. Lead PI: Shinoda (TAMUCC)
- **NSF (2014-2019, \$670,121):** Quantifying Predictability Limits, Uncertainties, Mechanisms, and Regional Impacts of Pacific Decadal Climate Variability, Institutional PI. Lead PI: Miller (SIO)
- **NOAA (2016-2019, \$180,000):** Improvement of MJO simulation in NCEP Coupled Forecast System: Upper ocean and air-sea coupled processes, Institutional PI. Lead PI: Shionda (TAMUCC)
- **NSF (2014-2019, \$500,000):** Decadal Variability in the North Atlantic Extra-Tropics: The Role of Coupling Between Atmospheric Blocking and the Atlantic Multidecadal Oscillation, co-PI. Lead PI: Ummenhofer (WHOI)
- **#ONR Young Investigator Award (2015-2019, \$490,482):** Dynamics, Impacts, and Predictability of Coupled Ocean-Atmosphere Interactions in the Northern Indian Ocean.
- **NASA (2015-2018, \$494,210):** Coupling Between the Atmospheric Intra-Seasonal Variability and Ocean Circulation in the Northern Hemisphere, co-I. PI: Kwon (WHOI)
- **#NSF (2014-2015, \$72,000):** Regional Coupled Ocean-Atmosphere Feedback Processes Affecting Climate Along the California Coast.
- **#ONR (2011-2014, \$91,000):** Coupled Ocean-Atmosphere Dynamics and Predictability of MJO's.

@ WHOI (until July 2024), Internal (competitive) Grants, **Secured ~\$495K**

- **#Fowler Ocean and Climate Fellow (2021-2023, \$140,213):** Western boundary currents and air-sea interactions: toward skillful predictive understanding and modeling of extreme storm and short-term climate events
- **#Independent Study Award (2019-2021, \$64,000):** Southern Ocean Eddies and Atmospheric Storm Tracks
- **#Ocean and Climate Change Institute (2015-2017, \$55,000):** A regional coupled model with explicit convection: diurnal rainfall and air-sea-land interaction in the Maritime Continent.
- **#Independent Study Award (2013-2015, \$49,000):** Submesoscale air-sea Interactions.

- **#Ocean and Climate Change Institute (2011-2013, \$140,000):** Dynamical analysis of surface wind responses to sea ice and surface temperature variations in the Arctic Ocean: Synthesis of data and model simulation. Co-PI: Yang (WHOI)
- **#Independent Study Award (2012-2013, \$47,000)** Trends in coastal upwelling and sea breeze in the California-Oregon Coast: Land-ocean-atmosphere interactions in data and model.

SELECTED SEMINARS AND TALKS

In the last 5 years: for the full list, see [here](#).

@ denotes presentations made as a UH faculty member

- @(Invited) Sea surface warming and mesoscale ocean-to-atmosphere feedback driven by large-scale offshore wind farms. Mesoscale and Sub-mesoscale Ocean Dynamics and Air-Sea Interaction. UH Manoa, 08/18/2025
- @(Invited) Wave-mediated air-sea flux parameterizations in non-equilibrium sea states and high winds. ONR MARBLES Workshop, Seoul, 06/25/2025
- (Invited) “Coupled” effects of offshore wind farms. BOEM Pacific Upwelling Workshop, UC Santa Cruz, 03/04/2025
- (Invited) Improving wave-mediated air-sea momentum flux parameterization; Geophysical Flows: From the Field to the Lab Discussion meeting (January 10 - 12, 2024), Chennai, India, 01/11/2024
- (Invited) Ocean Mesoscale Air-Sea Interactions: Physics, Impacts, and Role of Surface Waves, AOGS, Singapore, 08/04/2023
- (Invited) Surface waves, aerodynamic roughness & air-sea momentum flux: Mesoscale Air-Sea Interactions & Offshore Wind Energy, Yonsei University, Seoul, South Korea, 07/20/2023
- (Invited) Surface Waves, Aerodynamic Roughness, and Air-Sea Momentum Flux in the Context of Mesoscale and Frontal-Scale Air-Sea Interaction. Mid-latitude Ocean-Atmosphere Interactions: Their Processes and Predictability. Toyama, Japan, 06/16/2023
- Wind-Current Coupling in the Context of Mesoscale and Frontal-Scale Air-Sea Interaction. Odysea winds and currents Webinar, 06/06/2023
- (Invited) Surface Waves, Aerodynamic Roughness, and Air-Sea Momentum Flux, Japan Geosciences Union (JpGU), 05/21/2023.
- (Invited) Coupled Modeling of Mesoscale Air-Sea Interaction: Physics, Impacts, and Role of Surface Waves. Seoul National University, Korea, invited, 09/2022
- (Invited) Ocean Mesoscale Air-Sea Interaction over Gulf Stream: Drivers, Physics, and Influence. US CLIVAR With the Gulf Stream Workshop, 06/2022
- Mesoscale Air-Sea Interaction Working Group: Motivation, Science Goals, and Recommendations. US CLIVAR POS Panel Meeting, 08/21/2021
- (Invited) Impacts of ocean-atmosphere coupled feedback on weather and climate. NOAA Priorities for Weather Research (PWR) Observations and Data Assimilation Task Team Symposium. 07/08/2021
- (Invited) Impacts of ocean currents on the extratropical storm track through the “relative wind” effect. International Workshop on Midlatitude Air-Sea Interaction. 06/10/2021